

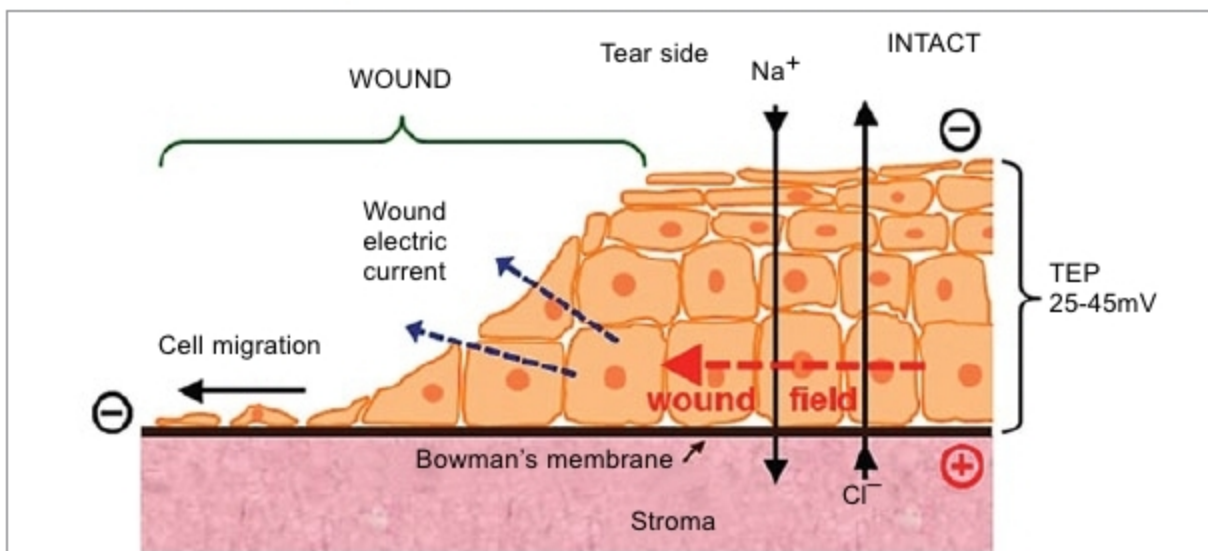


Fig. 3: Ionic basis of cornea wound's electrical response. The intact corneal epithelium (right) pumps sodium and chloride in opposite directions

exposed to nanogenerators.

Low-power pulses boost viability for a type of skin cell called fibroblasts, and exposure to the nanogenerator's pulses encourages the fibroblasts to line up, which is a crucial step in wound healing. This also produces more biochemical substances that promote tissue growth.

To power the electric bandage, a wearable nanogenerator is made by overlapping sheets of poly-tetra-fluoro-ethylene (PTFE), copper foil and polyethylene-terephthalate (PET). The nanogenerator converts skin movements, which occur during normal activity or while breathing, into small electrical pulses. **Current flows to the two working electrodes placed on either side of the skin wound to produce a weak electric field, which heals the wound within three days.** Scientists attribute faster wound healing to enhanced fibroblast migration, proliferation and differentiation induced by the electric field.

Mechanism of the wound healing process

When skin is injured, the body sets into motion an automatic series of events in four overlapping phases to repair the injured tissues.

Haemostasis phase. In this phase, healing starts at the onset of injury. The body activates its emergency repair system of blood clotting to stop the bleeding and forms a dam to block drainage. Meanwhile, platelets, which is one component of blood, start their activity. When there is a wound, platelets release mediators and growth factors for repair of tissues. An enzyme called thrombin is released that initiates the formation of a fibrin mesh, which strengthens platelet clumps into a stable clot.

Inflammatory phase. Bacteria is destroyed and debris is removed by neutrophils, a type of white blood cell that enters the wound. Neutrophils then leave, and specialised cells called macrophages arrive to continue scavenging work. These cells also secrete growth factors and proteins to facilitate tissue repair.

Proliferative phase. In this phase, shiny deep red granulation tissues fill the wound bed with connective tissue and new blood vessels are formed. Then, wound margins contract and pull towards the centre of the wound.

Next, epithelial cells (cells that line the body, skin, organs, etc) arise

from the wound bed or margins and begin to migrate across the wound bed in leapfrog fashion until the wound is covered with epithelium.

Remodelling phase. In this phase, new tissues remodel and slowly gain strength and flexibility.

In case of diabetes, due to uncontrolled high blood glucose level, ability of white blood cells to fight bacteria is greatly reduced. Also, blood circulation slows down, making it more labourious for the body to transfer nutrients to the wound site. That is why the wound takes longer to heal.

After a physical injury, in a moist wound environment, there is a lateral voltage gradient of 140mV/mm at the wound edge that decays to 10mV/mm at 500 to 1000 micrometres from the wound edge. In Fig. 3, it can be seen that the intact epithelium (right) pumps sodium and chloride ions in opposite directions to generate and maintain a trans-epithelial electrical potential difference (TEP) of approximately 45mV, which is necessary to maintain healthy skin.

Injury to the skin collapses the TEP locally at the wound to zero and significant electric fields build up around the wound (red arrow), which also generate large electric currents flowing out of the wound (blue arrows). Electric fields are intrinsically directional. These are oriented towards the wound, with the wound as a negative cathode.

Epithelial cells respond to the physiological strength of electric fields by migrating to the cathode, suggesting these use the natural, endogenous wound as the directional cue to migrate into the wound, thus enhancing healing.

Nanogenerator

A nanogenerator converts mechanical/thermal energy produced

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